

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Microprocessor. Compare Intel 8085, 8086 and 80386 Microprocessors on the basis of their features and internal architectures. 7
b) Explain flags in 8085 with the help of a diagram. What is the function of LDAXB? 8
2. a) Write an ALP in 8085 to find whether the given number is palindrome or not. 8
b) Write a 8085 program to find smallest of three numbers. 7
3. a) Define T-state. Draw the labeled timing diagram for the instruction LXI B, C00FH. 7
b) What are the three basic functions microprocessor should satisfy to interface any memory chip? Explain how 2048×8 EPROM is interfaced with 8085 microprocessor. 8
4. a) What are interrupts? Differentiate between I/O mapped I/O and Memory mapped I/O. 7
b) Draw and explain the functional block diagram of 8259A PIC. 8
5. a) Write an assembly language program for 8086 to find the sum of two numbers input by the user through the keyboard and display the sum on the screen. 8
b) Write an 8086 ALP for MASM in DOS-BIOS mode to display the string "Pokhara University" on screen without using 09H. 7
6. a) Explain procedures and macros. Why IVT is used in microprocessor. 7
b) Draw and explain the connections to show how RS-232C is made compatible with TTL logic along with its connections for null modem. 8
7. Write short notes on: (Any two) 2×5
 - a) HOLD and HLDA pin
 - b) BISYNC Protocol
 - c) DMA